Part 1c: More on 2nd-degree Price Discrimination

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Microeconomics 2 *(Module 1, 2024)*.

Suggested readings for this part

Readings in Class Textbooks:

- **Varian, 8th edition:**
  - Ch 25: Monopoly Behavior
- **Carlton and Perloff:**
  - Ch 9: Price Discrimination
  - Ch 10: Advanced Topics in Pricing

Bundling (Tie-in Sales)

Setting

- Two goods, A and B. Let's assume zero cost of production for simplicity.

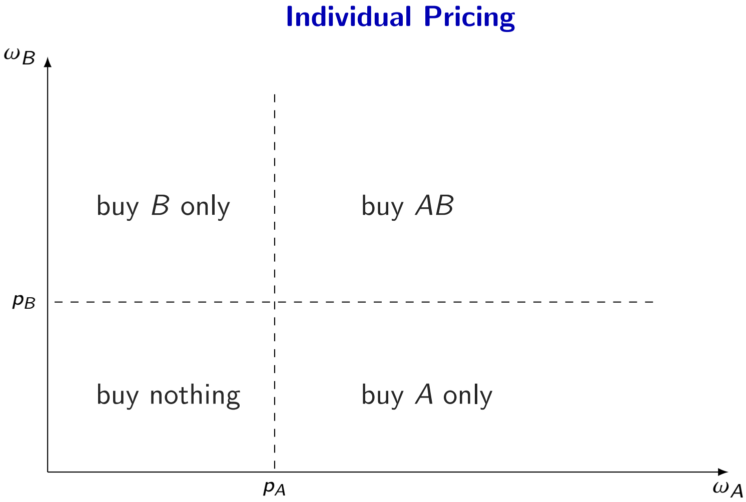
- Multiple consumers with **independent unit demand** for each product described by its willingness to pay (wtp):

$$\omega_A \text{ for good } A \qquad \text{and} \qquad \omega_B \text{ for good } B$$

- We will consider three pricing regimes for the firm:
  - **Individual Pricing**: p_A and p_B.
  - **Pure Bundling:** p_{AB} (price of the bundle).
  - **Mixed Bundling:** p_A, p_B and p_{AB} such that

$$p_{AB} \leq p_A + p_B$$

How do consumers decide what to buy?



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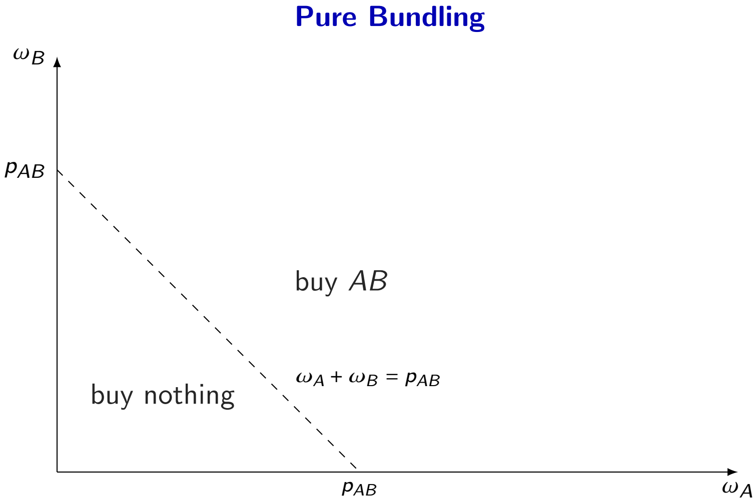

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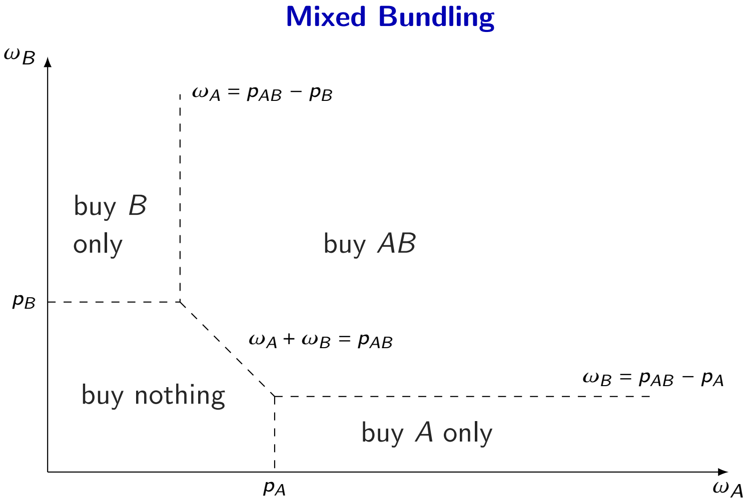


Which one is better for the firm?

- Three pricing regimes:
  - **Individual Pricing**: p_A and p_B.
  - **Pure Bundling:** p_{AB} (price of a bundle).
  - **Mixed Bundling:** p_A, p_B and p_{AB} such that

    $$p_{AB} \leq p_A + p_B$$

- What is the profit-maximizing pricing regime? Depends
  - a comparison between **individual pricing** and **pure bundling** is ambiguous (depends on preferences)
  - **mixed bundling** is less restricted, so it can never be worse than **individual pricing** or **pure bundling**.
  - **mixed bundling** can be strictly better sometimes (depends on preferences).

Example

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo			
Juliet			

- Romeo and Juliet are on a date in a restaurant

Example

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo			
Juliet			

- Romeo and Juliet are on a date in a restaurant
- Assumptions:
  - everyone pays for his or her meal
  - no food sharing (you cannot eat half of something)
  - no resale of any kind

Example - I

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	20	100
Juliet	70	30	100

Example - I

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	20	100
Juliet	70	30	100

- different relative tastes for components
- but similar value for the whole meal
- here, **pure bundling** is better than **individual pricing**
- what about **mixed bundling**?

Example - II

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	10	90
Juliet	10	40	50

Example - II

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	10	90
Juliet	10	40	50

- different value for the whole meal
- and very different relative tastes for components
- here, **individual pricing** is better than **pure bundling**
- but we can do even better by using **mixed bundling** here

Example - III

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	30	110
Juliet	70	10	80

Example - III

	Steak (A)	*Cake (B)*	*Meal (AB)*
Romeo	80	30	110
Juliet	70	10	80

- similar relative tastes for components
- but different value for the whole meal
- here, **individual pricing** is better than **pure bundling**
- what about **mixed bundling**?

Vertical Differentiation

Remainder: Price Discrimination with Unit Demands

- Consider a **unit demand setting** with a homogeneous good
- We have concluded (see Part 1b) that the **2nd-degree PD is equivalent to the nondiscriminatory linear pricing**. In other words, allowing for quantity-specific prices cannot increase monopoly's profit in this setting.
- We also argued that the monopoly can differentiate its products and price them differently when consumer specific prices are not feasible or illegal. Under certain conditions, differentiation increases monopoly's profit.

Product Differentiation

A monopoly can increase its market power and market share by differentiating products:

- Vertical Differentiation — different quality.
  - if prices are the same, people only buy the highest quality product.
  - **Examples:** *airplane tickets, trim levels for cars*
- Horizontal Differentiation — different features, similar quality.
  - if prices are the same, people buy different products.
  - **Examples:** *taste, color*

Example

- A monopolist has two units of the good (already produced) and values them at zero.
- Suppose that there are two unit-demand customers with

$$\omega_1 = 10 \qquad \text{and} \qquad \omega_2 = 4$$

- Without consumer-specific prices, **the best option is to sell one unit for 10**.
- Let's change "*the quality*" of one item, effectively differentiating them into two varieties (*High quality and Low quality*).
- Now we can **price them individually** since, technically, they are different products that can be priced differently.
- This is called vertical differentiation.

Model

- Good is offered in two varieties — high and low quality
- The monopolist has one unit of each variety and has no value from keeping any items.
- There are two consumers ($N = 2$) with unit demands. Their **wtps** are as follows:

	High	*Low*
Consumer 1	ω_1^H	ω_1^L
Consumer 2	ω_2^H	ω_2^L

- Let's assume that

$$\begin{array}{rl} (i) & \omega_1^H > \omega_1^L \geq 0 \\ (ii) & \omega_2^H > \omega_2^L \geq 0 \\ (iii) & \omega_1^H - \omega_1^L > \omega_2^H - \omega_2^L \end{array}$$

Social Optimum

- To be concrete, let's assume the following values:

	High	*Low*
Consumer 1	$\omega_1^H = 10$	$\omega_1^L = 5$
Consumer 2	$\omega_2^H = 4$	$\omega_2^L = 3$

- The social optimum is given by
  - *High* good $\Rightarrow$ Consumer 1
  - *Low* good $\Rightarrow$ Consumer 2

  This is due to assumption (iii) $\omega_1^H - \omega_1^L > \omega_2^H - \omega_2^L$.

- The social welfare in this market

$$SW = \omega_1^H + \omega_2^L = 13$$

Price Discrimination

- Uniform price (same for both varieties)
  - C1 receives H for $p = 10$.
  - $PS = 10,\ CS = 0,\ SW = 10,\ DWL = 3$.

- 1st or 3rd degree PD (consumer specific prices)
  - C1 gets H for $p_H = 10$, and C2 gets L for $p_L = 3$.
  - $PS = 13,\ CS = 0,\ SW = 13,\ DWL = 0$.

- Will $p_H = 10,\ p_L = 3$ work without consumer discrimination?

Price Discrimination

- Uniform price (same for both varieties)
  - C1 receives H for $p = 10$.
  - $PS = 10, \; CS = 0, \; SW = 10, \; DWL = 3$.
- 1st or 3rd degree PD (consumer specific prices)
  - C1 gets H for $p_H = 10$, and C2 gets L for $p_L = 3$.
  - $PS = 13, \; CS = 0, \; SW = 13, \; DWL = 0$.
- Will $p_H = 10, \; p_L = 3$ work without consumer discrimination?
  - No! Consumer 1 will buy *Low* instead (a surplus of 2)
  - **What are the right prices here?**

Price Discrimination

- Uniform price (same for both varieties)
  - C1 receives H for $p = 10$.
  - $PS = 10,\ CS = 0,\ SW = 10,\ DWL = 3.$
- 1st or 3rd degree PD (consumer specific prices)
  - C1 gets H for $p_H = 10$, and C2 gets L for $p_L = 3$.
  - $PS = 13,\ CS = 0,\ SW = 13,\ DWL = 0.$
- Will $p_H = 10,\ p_L = 3$ work without consumer discrimination?
  - No! Consumer 1 will buy *Low* instead (a surplus of 2)
  - **What are the right prices here?**
  - We can argue "informally" that the profit-maximizing price combination is

$$p_H = 8 \qquad \text{and} \qquad p_L = 3$$

What is the best way to sell H to 1 and L to 2?

Choose p_H and p_L to

$$\max_{p_H, p_L} \quad p_H + p_L$$

such that:

Consumer 1 wants to buy H	$10 - p_H \geq 0$	(1)
Consumer 2 wants to buy L	$3 - p_L \geq 0$	(2)
Consumer 1 wants to buy H instead of L	$10 - p_H \geq 5 - p_L$	(3)
Consumer 2 wants to buy L instead of H	$3 - p_L \geq 4 - p_H$	(4)

Note that

$$10 - p_H \geq 5 - p_L \quad \Longrightarrow \quad p_H - p_L \leq 5$$
$$3 - p_L \geq 4 - p_H \quad \Longrightarrow \quad p_H - p_L \geq 1$$

Analytical Solution to 2nd-degree PD

- Let's simplify the system of constraints
- (1) is redundant because of (3) and (2)

$$10 \; - \; p_H \; \geq \; 5 \; - \; p_L \; > \; 3 \; - \; p_L \; \geq \; 0$$

- Then (2) must bind , i.e.,

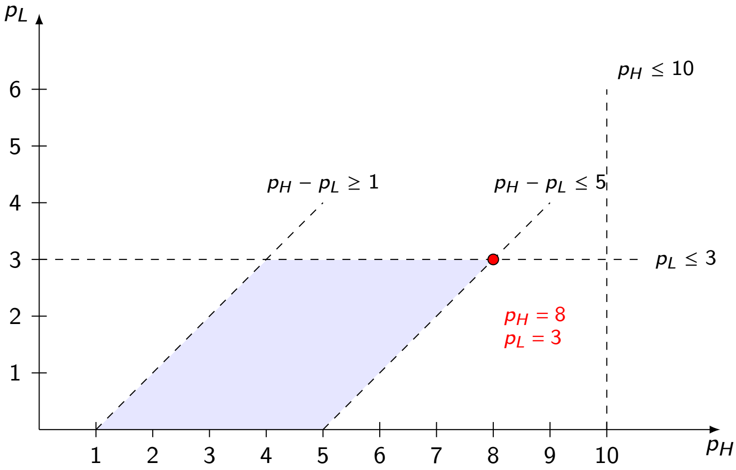
$$\boxed{p_L \; = \; 3}$$

- Finally, (3) must bind, i.e.,

$$10 \; - \; p_H \; = \; 5 \; - \; p_L \qquad \Rightarrow \qquad \boxed{p_H \; = \; 8}$$

- Then $PS = 11$, $CS_1 = 2$, $CS_2 = 0$, $SW = 13$, $DWL = 0$.

Alternative Graphical Solution



Summary on Vertical Differentiation

- Typically consumers have differentiated tastes for varieties — some customers might really dislike the "low" quality items.
- Individual pricing of varieties allows the monopoly to extract surplus of variety-loving consumers without using consumer specific prices.
- Thus, infeasible consumer specific prices are effectively replaced with feasible variety specific prices.

Vertical Differentiation with Quality Choice

Quality Choice for a Monopoly

- Before selling its products, a monopoly can design products with specific quality levels
- All consumers prefer higher quality, but have a different willingness to pay (wtp) for quality.
- When consumers' tastes are different, it is optimal for the monopoly to engage in vertical differentiation

General Model

- There are two types of buyers: Premium and Regular. All customers have a unit demand for a good..
- There are N customers in total, $\lambda \in [0, N]$ of them are Regular.
- A buyer's monetary value for the good depends on the good's **quality** $q \geq 0$ and is different for the two types:
  - value $v_P(q)$ for Premium (with $mv_P(q)$ for marginal value);
  - value $v_R(q)$ for Regular (with $mv_R(q)$ for marginal value);
- Both types like quality, but *Premium* customers like quality more than *Regular*, so

$$mv_F(q) \; > \; mv_R(q) \; > \; 0 \qquad \forall q \geq 0$$

- The per-unit production cost for a good with quality q is given by $c(q) \;\; (c'(q) > 0, \; c''(q) > 0)$. Note that q stands for **quality** here, not **quantity**.
- This problem can be solved in the general form, but it is too intense mathematically. **We will solve a simplified version.** .

Simplified Model

- Consider a market for a new iPhone. There are 100 customers in total, $\lambda \in [0, 100]$ of them are Fans and $100 - \lambda$ are Regular.

- Suppose that value functions for iPhone's **quality** *(e.g., memory, speed, battery)* by Fans and Regular are

$$v_F(q) \ = \ 6q \qquad \text{and} \qquad v_R(q) \ = \ 4q$$

- The per-unit production cost for **quality** q is given by $c(q) \ = \ q^2/2$

- The socially optimal phone parameters are: *(from $mv \ = \ mc$)*

$$\begin{array}{lllll} \textit{phone for Fans} & q_F^* = 6 \quad \Rightarrow & c_F = 18 & v_F = 36 & (\textit{“iPhone Pro”}) \\ \textit{phone for Rest} & q_R^* = 4 \quad \Rightarrow & c_R = 8 & v_R = 16 & (\textit{“iPhone”}) \end{array}$$

- The social welfare *(in money)* is given by

$$\lambda[v_F - c_F] \ + \ (100 - \lambda)[v_R - c_R] \ = \ 800 \ + \ 10\lambda$$

1st or 3rd Degree Price Discrimination

- If Apple can charge consumer specific prices, then it will be able to extract full surplus
- For Fans, it will offer the iPhone Pro model at

$$(q_F,\ p_F)\ =\ (6,\ 36)$$

- For Regular, it will offer the iPhone model at

$$(q_R,\ p_R)\ =\ (4,\ 16)$$

- Apple's profit in this case is

$$\lambda[p_F - c(q_F)]\ +\ (100-\lambda)[p_R - c(q_R)]\ =\ 800\ +\ 10\lambda$$

Naive Solution to 2nd-degree PD

- Let's simply take the quality levels from the social optimum and apply vertical differentiation logic to prices.
- The socially efficient quality levels at

$$q_F = 6 \qquad \text{and} \qquad q_R = 4$$

- Apple charges **Regular** customers to extract all surplus

$$p_R = 4 \cdot q_R = 16$$

- And Apple charges **Fans** to make them indifferent between two models, i.e.,

$$p_F = 16 + 12 = 28$$

- Apple's profit in this case is

$$\lambda[28 - 18] + (100 - \lambda)[16 - 8] = 800 + 2\lambda$$

2nd-degree PD Problem

Apple designs a phone-price bundle for each type

$$(q_F,\ p_F) \qquad \text{and} \qquad (q_R,\ p_R)$$

to maximize its profit

$$\max_{(q_F,p_F),(q_R,p_R)} \quad \lambda(p_F - c(q_F)) + (100-\lambda)(p_R - c(q_R))$$

subject to **participation** and **self-selection constraints**:

$$
\begin{aligned}
&\textit{Fans buy iPhone Pro:} & 6\,q_F - p_F &\geq 0 & (1)\\
&\textit{Rest buy iPhone:} & 4\,q_R - p_R &\geq 0 & (2)\\
&\textit{Fans prefer iPhone Pro:} & 6\,q_F - p_F &\geq 6\,q_R - p_R & (3)\\
&\textit{Rest prefer iPhone:} & 4\,q_R - p_R &\geq 4\,q_F - p_F & (4)
\end{aligned}
$$

Correct Solution to 2nd-degree PD

- Apple will offer

$$(q_F,\ p_F) \qquad \text{and} \qquad (q_R,\ p_R)$$

- Using the standard *Vertical Differentiation* logic, constraints (2) and (3) must bind, and we have

$$p_R \ = \ 4q_R \qquad \text{and} \qquad p_F \ = \ 6q_F \ - \ 2q_R$$

- Apple picks quality levels by maximizing

$$\max_{q_F, q_R} \quad \lambda(6q_F - 2q_R - c(q_F)) \ + \ (100 - \lambda)(4q_R - c(q_R))$$

Correct Solution to 2nd-degree PD

- The optimal quality for the iPhone Pro is

$$q_F \;=\; 6 \qquad \textit{(socially optimal)}$$

- The optimal quality for the iPhone is

$$q_R \;=\; 4 \;-\; 2\,\frac{\lambda}{100-\lambda} \qquad \textit{(less than socially optimal)}$$

- This solution makes sense only when

$$q_R \;\geq\; 0 \qquad \Rightarrow \qquad \boxed{\lambda \;\leq\; 200/3 \;\approx\; 66.66}$$

- Apple's profit is given by

$$\lambda[36 - 2q_R - 18] \;+\; (100-\lambda)[4q_R - 0.5q_R^2]$$

Correct Solution to 2nd-degree PD

- If

$$\lambda \;>\; 200/3 \;\approx\; 66.66$$

Apple should offer

$$(q_F,\; p_F) \qquad \text{and} \qquad (0,\; 0)$$

where

$$q_F \;=\; 6 \qquad \text{and} \qquad p_F \;=\; 36$$

- In other words, Apple only serves Fans and extracts all surplus from them
- Apple's profit in this case is given by

$$\lambda[36-18] \;+\; (100-\lambda)[0-0] \;=18\lambda$$

Example: Equal Shares $(\lambda = 50)$

- The maximal social welfare is given by

$$800 + 10\lambda = 1400$$

- The naive solution consists of offerings

$$(6,\ 28) \qquad \text{and} \qquad (4,\ 16)$$

  for a profit of

$$\lambda[28 - 18] + (100 - \lambda)[16 - 8] = 800 + 2\lambda = 900$$

- The correct solution consists of offerings

$$(6,\ 32) \qquad \text{and} \qquad (2,\ 8)$$

  for a profit of

$$\lambda * [32 - 18] + (100 - \lambda) * [8 - 2] = 600 + 8\lambda = 1000$$

Example: More Fans $(\lambda = 80)$

- The maximal social welfare is given by

$$800 + 10\lambda = 1600$$

- The naive solution consists of offerings

$$(6,\ 28) \qquad \text{and} \qquad (4,\ 16)$$

for a profit of

$$\lambda[28-18] + (100-\lambda)[16-8] = 800 + 2\lambda = 960$$

- The correct solution consists of offerings

$$(6,\ 36) \qquad \text{and} \qquad (0,\ 0)$$

for a profit of

$$\lambda * [36-18] + (100-\lambda) * [0-0] = 18\lambda = 1440$$

Checklist for Part 1c

1. **Bundling:** Individual Pricing vs. Pure Bundling vs. Mixed Bundling.
2. **2nd-degree PD** with vertical differentiation.
3. **2nd-degree PD** with vertical differentiation and quality choice.

Play These Two Games for Next Time

Grade Game

Each student have to pick between two options: A or B.

I will randomly pair you with another student in the class. Neither you nor your pair will ever know with whom you were paired. We will be using the **HSE 10-point grading system**. Here is how grades may be assigned for this class (hypothetically).

- If you put **A** and your pair puts **B**, then you will get **9**, and your pair will get **4** .
- If both of you put **A**, then you both will get grade **5**.
- If you put **B** and your pair puts **A**, then you will get grade **4**, and your pair will get grade **9**.
- If both of you put **B**, then you will both get grade **7**.

Make your choice

Guess 2/3 of the Average Game

Rules of the game:

Each of you have to choose an integer between 1 and 100 in order to guess *"2/3 of the average of the responses given by all students in the class"*. Each student who guesses the integer closest to the 2/3 of the average of all the responses, wins.

Example: (with only 3 students)

Choices made:	42, 13, 80
The average:	45
2/3 of the Average	30
The winner	Student who wrote 42

What is your guess?